# Ethan Mahajan

ENGINEERING PORTFOLIO

Mechanical engineering student focused on CAD, manufacturing, prototyping, and hands-on problem solving.

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### From working prototypes to current design work

These five projects connect CAD, fabrication, electronics, and engineering communication. Each summary identifies my contribution, the evidence available, and the work still pending. The linked case studies provide complete documentation and demonstrations.

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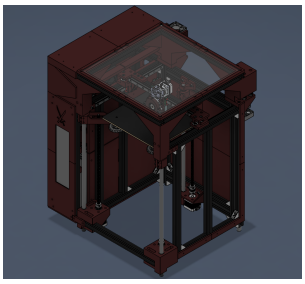
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## Artorius: FDM 3D Printer

Operational; refinement ongoing | Independent design and build



Revised CAD assembly. The linked case study includes footage of the operational printer.

**Design and contribution.** Recurring problems with an older BIQU B1 led me to redesign its architecture around reusable motors and a 24 V supply. I developed a CoreXY frame, motion system, and custom toolhead in CAD, fabricated fixtures, assembled the machine, and integrated CAN-bus electronics with a serviceable enclosure.

**Evidence and next steps.** The printer is operational and the linked video shows Benchy printing. Motion tests explored approximately 1,000 mm/s; this is not a validated print-quality speed. Extra frame support did not fully resolve resonance. Alignment, packaging, vibration, and repeatable print quality remain refinement priorities.

**Tools:** CAD, FDM fabrication, CoreXY motion, CAN bus

Full case study and supporting evidence

## Phone Mount

Completed; usable first prototype | Independent design and fabrication



The usable first prototype: adjustable height, arm rotation, and phone orientation.

**Design and contribution.** I needed an inexpensive, adjustable way to film engineering builds with an iPhone. I modeled Fusion 360 fixtures for height, arm rotation, and phone orientation, printed them in ABS, and assembled the mount with aluminum extrusion, heat-set inserts, and fasteners. Chamfers and clearances supported FDM fabrication and assembly.

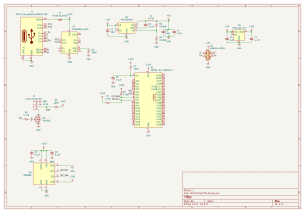
**Evidence and next steps.** The usable first prototype cost approximately $18 including extrusion, or $4 excluding it, as estimated in the project video. It became my project filming mount. Assembly exposed a missing screw-access hole; holder play and base shake suggest future stiffness improvements. A second build was unnecessary for the intended use.

**Tools:** Fusion 360, OrcaSlicer, ABS printing

Full case study and supporting evidence

## Air Purifier

Design in progress | Independent mechanical and electronics design



Completed KiCad schematic; board layout and physical validation are pending.

**Design and contribution.** A dorm-room need for filtration and background noise motivated a compact purifier concept using a repurposed 120 mm fan. The planned assembly combines a printed housing, filter, particulate-matter sensor, and custom controller. I completed the electrical schematics in KiCad; PCB placement has not started.

**Evidence and next steps.** The demonstrated deliverable is schematic work, not a functioning purifier. Next steps are board layout, housing and board fabrication, and system integration. Airflow, fan control, sensor response, noise, and proposed filter-replacement detection still require physical testing.

**Tools:** KiCad, CAD, planned FDM fabrication

Full case study and supporting evidence

## Shape Machine

Contribution ongoing | Website content and documentation

Shape Computation Lab identity. The linked page distinguishes my role from the team's technology.

**Design and contribution.** Within an interdisciplinary team, I improve the existing Shape Computation Lab website through content review, error correction, template maintenance, and GitHub documentation in the team’s enterprise repository. The goal is to make explanations of Shape Machine's specialized shape-computation technology clearer and more consistent for visitors with varied technical backgrounds.

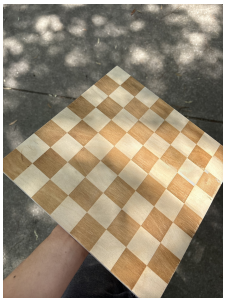
**Evidence and next steps.** My contribution is ongoing and focuses on the website's existing content and presentation. The case study describes those responsibilities and links to demonstrations of the team's technology for context. Specific before-and-after contribution examples remain to be documented; I do not claim authorship of the underlying technology.

**Tools:** Content editing, templates, GitHub

Full case study and supporting evidence

## Laser-Engraved Chess Board

Physical board documented; metal pieces pending | Independent design and fabrication



Physical laser-engraved board; the CNC-machined metal pieces have not been made.

**Design and contribution.** I am developing a coordinated chess set through Georgia Tech's Invention Studio, combining a laser-engraved board with planned CNC-machined metal pieces. The CAD design prioritizes a manufacturable playing surface and stable, low-center-of-mass pieces. Separating the board and piece stages lets fabrication progress incrementally.

**Evidence and next steps.** Photographs document the physical board’s engraved playing surface and personalized reverse; a video records the laser engraving process. The metal pieces have not been made and the full set is pending. Developing the CNC manufacturing approach is the next stage.

**Tools:** CAD, laser engraving; planned CNC machining

Full case study and supporting evidence